Exp.No: 14 Packet Sniffing and Raw

Aim:

To write a program using RAW sockets to implement packet sniffing.

ALGORITHM:

1. Create a raw socket
2. Bind the socket to network ~~top~~ interface
3. Receive network packets using RAW socket.
4. Extract packet header info
5. Display the source IP, destination IP and protocol
6. Continue capturing packet and display the details

Code:

```
import socket
import struct
S= socket.socket (socket.AF_INET, socket.SOCK.RAW,
                  socket.IPPROTO_IP)
print ("Packet Sniffer started...")
while True:
    packet, addr = S.recvfrom(65565)
    ip_header = packt[0:20]
    iph = struct.unpack('!BBHHHBBH4S4s', ipheader)
    Source_ip = socket.inet_ntoa (iph[8])
    destination_ip = socket.inet_ntoa (iph[9])
    Protocol = iph[6]
    print ("Source IP:", source_ip)
    print ("Destination IP:", destination_ip)
```

print ("Protocol:", protocol)
print ("-" * 40)

INPUT:

Network packets transmitted through the network

Output:

Packet sniffer started

Source IP : 192.168.1.10

Destination IP : 142.250.183.14

Protocol : 6

Source IP : 192.168.1.10

Destination IP : 142.250.72.14

Protocol : 17

Result:

Thus the program to implement packet sniffing using RAW sockets was executed successfully